Task B1 — Constraint Triangle Application

LogiEdge: Edge AI cold-chain monitoring | FreightBridge Logistics Pvt. Ltd. | Component B, Module 2

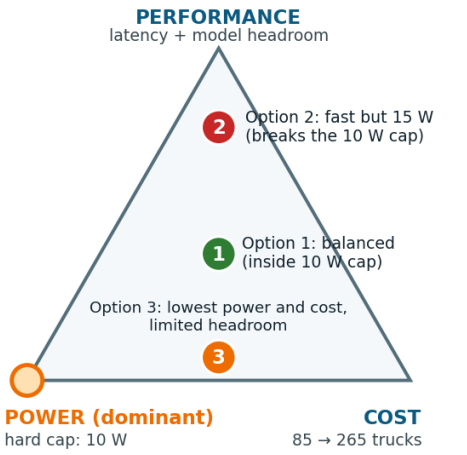
At a glance

Question	Answer
Dominant constraint vertex	Power: the 10 W AI budget on the 12 V truck supply is a hard physical cap
Recommended option	Option 1: Raspberry Pi 5 (8 GB) + AI HAT+ (13 TOPS)
Rejected options	Option 2 breaks the 10 W cap and costs 3×. Option 3 has no headroom and needs custom engineering.
Hardware cost (Option 1)	₹12.75 lakh for 85 trucks; ₹39.75 lakh for 265 trucks

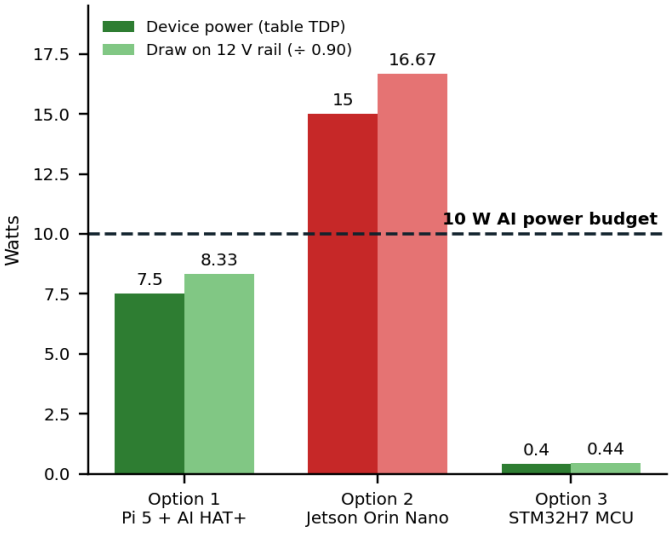
1 The constraint triangle for LogiEdge

Vertex	What it means here	Requirement	Metric used
Performance	Meets the detection deadline and leaves model headroom	Alert within 90 s of a fault	Detection-to-alarm time; compute headroom
Power	Fits the truck's electrical supply	10 W AI budget; 12 V supply via DC-DC converter	Device power and draw on the 12 V rail
Cost	Affordable across the fleet	85 trucks in the pilot; 265 at full scale	Unit price × fleet size

Constraint triangle (schematic placement)



Power against the 10 W budget



2 Applying the triangle to each option

2.1 Performance: the 90 s latency requirement

Option	Inference time (assumed)	Detection-to-alarm	Share of 90 s	Peak compute used
1 Pi 5 + AI HAT+	≈ 0.01 s	10.15 s	11.3%	0.0015% of 13 TOPS
2 Jetson Orin Nano	< 0.01 s	10.15 s	11.3%	0.0003% of 67 TOPS
3 STM32H7 MCU	≈ 0.02 s	10.15 s	11.3%	40% of ≈ 0.5 GOPS

Finding: all three alert in about 10.15 s (10 s window + 0.05 s inference allowance + 0.1 s alarm, as in Task A1), leaving 79.85 s spare. Latency does not separate the options; the window length dominates, not the processor.

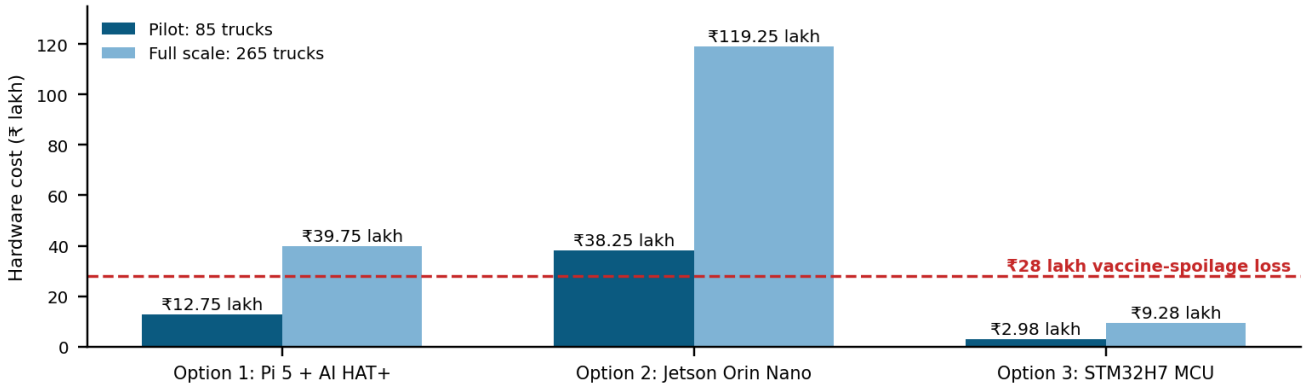
2.2 Power: the 10 W AI budget on a 12 V supply

Option	Device power	Draw on 12 V rail (÷ 0.90)	Current at 12 V	Share of 10 W (device / rail)	Energy per day (24 h, rail)
1 Pi 5 + AI HAT+	7.5 W	8.33 W	0.69 A	75% / 83%	200 Wh
2 Jetson Orin Nano	15 W	16.67 W	1.39 A	150% / 167%	400 Wh
3 STM32H7 MCU	0.4 W	0.44 W	0.04 A	4% / 4%	10.7 Wh

Finding: Option 1 passes whether the 10 W limit applies at the device (2.5 W spare) or on the 12 V rail (1.67 W spare). Option 2 fails under both readings.

2.3 Cost: 85-truck pilot and 265-truck scale-up

Option	Per truck	Pilot (85 trucks)	Full scale (265 trucks)	Relative to Option 1
1 Pi 5 + AI HAT+	₹15,000	₹12,75,000 (₹12.75 lakh)	₹39,75,000 (₹39.75 lakh)	1.0×
2 Jetson Orin Nano	₹45,000	₹38,25,000 (₹38.25 lakh)	₹1,19,25,000 (₹1.19 crore)	3.0×
3 STM32H7 MCU	₹3,500	₹2,97,500 (₹2.98 lakh)	₹9,27,500 (₹9.28 lakh)	0.23×



Prices are the hardware figures given in the brief. They exclude enclosure, DC-DC converter, modem, installation and any custom-board engineering.

3 Dominant constraint vertex: Power

- **Latency does not decide it:** every option alerts in about 10.15 s against a 90 s limit.
- **Power is pass or fail:** Option 2 exceeds the 10 W cap by 50% at the device (15 W) and 67% on the rail (16.67 W). No software setting changes the 15 W figure given for moderate load.
- **Power also drives running cost:** energy per day, heat inside a sealed truck enclosure, and DC-DC converter sizing all scale with it. At 24 h operation Option 1 uses 200 Wh/day against 400 Wh/day for Option 2.
- **Cost breaks the tie:** between the two options that pass the power gate, cost and integration effort separate them.

Conclusion: Power is the dominant vertex. It is the only vertex that removes an option on a physical limit rather than a preference. Cost then separates the survivors, and performance is satisfied by all three.

4 Recommendation: argue for Option 1, against Options 2 and 3

Option	Verdict	Against the 90 s latency requirement	Against the 10 W budget	Against fleet-scale cost
1 Pi 5 + AI HAT+	**Choose**	10.15 s, 11% of the budget; 13 TOPS is vast headroom	7.5 W device (75%); 8.33 W on the rail (83%); passes both readings	₹12.75 lakh pilot, about 46% of one ₹28 lakh spoilage loss; ₹39.75 lakh at 265
2 Jetson Orin Nano	**Reject**	Also meets 90 s, but 67 TOPS is wasted on a model needing about 0.2 GOPS	15 W is 150% of the budget (167% on the rail): fails	₹38.25 lakh pilot, 137% of one ₹28 lakh loss; ₹1.19 crore at 265, 3× Option 1
3 STM32H7 MCU	**Reject (narrowly)**	Meets 90 s: about 0.02 s per inference, so compute is sufficient	Best at 0.4 W (4%)	Cheapest: ₹2.98 lakh pilot, ₹9.28 lakh at 265, before custom-engineering cost

Why Option 1

- It meets the 90 s and 10 W requirements with margin and costs about a third of Option 2 per truck.
- It runs Linux, so the Task A2 design (local MQTT broker, encrypted hash-chained alert log, TLS 1.3 uplink, sync agent, over-the-air updates) uses standard software instead of custom firmware.
- 13 TOPS leaves room for larger models when the system scales to all 265 trucks, including engine-anomaly detection, since the two earlier breakdowns were visible in sensor data three days before.

Why not Option 2

- It fails the hard power cap. NVIDIA lists a 7 W reference mode for the Orin Nano 8 GB (the original modes are 7 W and 15 W), which could fit under 10 W at reduced performance; NVIDIA's table for the Super software lists 15 W, 25 W and MAXN SUPER instead, so check the 7 W mode is available on your kit. Even so, the cost stays at ₹45,000 per truck and the extra 67 TOPS would go unused.
- It is a developer kit, which is not built for vehicle vibration and temperature.

Why not Option 3

- It is technically capable of today's classifier, so the rejection is about integration, not speed. It has about 1 MB of RAM and no Linux, so the local MQTT broker, encrypted log, TLS and updates would need custom firmware, changing the Task A2 design.
- Custom hardware carries one-off engineering cost (board design, firmware, vehicle testing) that is not in the ₹3,500. Option 3 saves ₹9.78 lakh at 85 trucks and ₹30.48 lakh at 265 trucks against Option 1, so it wins on cost only if that engineering stays below those amounts.
- It has little headroom for larger models at scale.

5 Risks of Option 1 and mitigations

Risk	Why it matters	Mitigation
Thin power margin	1.67 W spare on the rail; modem and sensors add load	Measure power under full load with the HAT+ running; confirm whether the modem sits outside the 10 W
Not automotive-rated	Vibration, heat, dust and supply spikes	Industrial enclosure, automotive-grade 12 V DC-DC converter with surge protection, industrial storage instead of an SD card
Accelerator compatibility	Vibration models may use operations the Hailo-8L does not support	Design the model for the accelerator, or run it on the Pi 5 CPU, since the workload is tiny
Parked-truck battery drain	200 Wh/day is about 17% of an assumed 1,200 Wh battery	Low-voltage cut-off and a low-power idle mode when the engine is off

6 Items to confirm

- The three vertices used here are Performance, Power and Cost. If your module defines the triangle differently (for example accuracy, latency and power), tell me and I will remap it.
- Whether the 10 W budget applies at the device or on the 12 V rail (both are shown; Option 1 passes either way).
- DC-DC efficiency (90% assumed) and whether the 4G modem and sensors sit inside the 10 W.
- Model size (5 million multiply-accumulates assumed); a much larger model reduces Option 3's headroom.
- Battery capacity (12 V, 100 Ah assumed) and whether the node stays powered when the engine is off.
- Any quoted engineering cost for the custom MCU board, to test the Option 3 break-even.

Appendix B1 — Assumptions and Workings

Decimal units. “Assumed” values are engineering estimates, not measurements.

A. Assumptions

Item	Value
DC-DC converter efficiency	90% (assumed)
Operating time	24 h per day (assumed)
Model workload	5 million multiply-accumulates = 10 million operations per 10 s window (assumed)
Inference deadline	0.05 s allowance, as in Task A1
Cortex-M7 effective throughput	≈ 0.5 GOPS with optimised int8 kernels (assumed)
Battery	12 V, 100 Ah = 1,200 Wh (assumed)
Prices, device power, TOPS	As given in the brief
Spoilage loss for comparison	₹28 lakh (from the brief)

B. Workings

Step	Calculation	Result
Detection-to-alarm	10 + 0.05 + 0.1	10.15 s; 10.15 ÷ 90 = 11.3%; spare 90 − 10.15 = 79.85 s
Peak compute needed	10 million operations ÷ 0.05 s	0.2 GOPS
Utilisation, Option 1	0.2 GOPS ÷ 13 TOPS	0.0015%
Utilisation, Option 2	0.2 GOPS ÷ 67 TOPS	0.0003%
Option 3 inference time	10 million ÷ 0.5 GOPS; utilisation 0.2 ÷ 0.5	0.02 s; 40%
Rail draw, Option 1	7.5 ÷ 0.90	8.33 W; 0.69 A at 12 V; 200 Wh/day
Rail draw, Option 2	15 ÷ 0.90	16.67 W; 1.39 A at 12 V; 400 Wh/day
Rail draw, Option 3	0.4 ÷ 0.90	0.44 W; 0.04 A at 12 V; 10.7 Wh/day
Share of 10 W, Option 1	7.5 ÷ 10; 8.33 ÷ 10	75%; 83%
Share of 10 W, Option 2	15 ÷ 10; 16.67 ÷ 10	150%; 167%
Battery share per day	200, 400, 10.7 ÷ 1,200 Wh	16.7%; 33.3%; 0.9%
Option 1 cost	15,000 × 85; 15,000 × 265	₹12,75,000; ₹39,75,000
Option 2 cost	45,000 × 85; 45,000 × 265	₹38,25,000; ₹1,19,25,000
Option 3 cost	3,500 × 85; 3,500 × 265	₹2,97,500; ₹9,27,500
Per-truck ratio	45,000 ÷ 15,000; 3,500 ÷ 15,000	3.0×; 0.23×
Pilot cost vs one loss	12.75 ÷ 28; 38.25 ÷ 28	46%; 137%
Option 3 break-even	(15,000 − 3,500) × 85; × 265	₹9,77,500 (₹9.78 lakh); ₹30,47,500 (₹30.48 lakh)